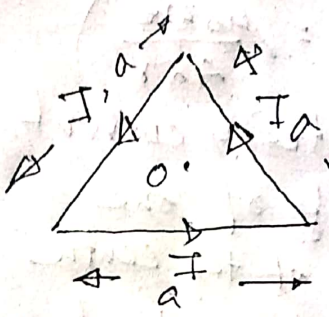
Problem on Magnetism 1 & 2

1. A proton and a alfa particle enter a magnetic field of constant magnitude with their velocities being ⊥ to the magnetic field. Find ratio of radii of their circular paths if they enter with (i) same velocity (ii) same linear momentum (iii) same kinetic energy.

[1:2, 2:1, 1:1]

2. 

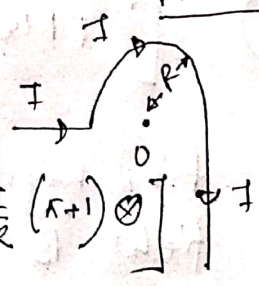
Find magnetic field at centroid O of a current loop (current I), which has the shape of an equilateral triangle of side 'a'.

$\left[\frac{\mu_0}{4\pi} \cdot \frac{18 I}{a} \right]$

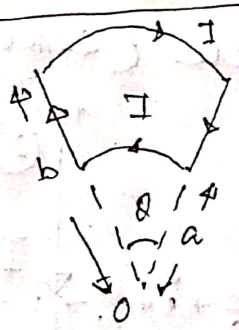
3. An electron enters an electric field $\vec{E} = (3\hat{i} + 6\hat{j} + 2\hat{k})$ v/m and magnetic field $\vec{B} = (2\hat{i} + 3\hat{j})$ T with a velocity $\vec{V} = (2\hat{i} + 3\hat{j})$ m/s. Find magnitude of force on electron ($e = 1.6 \times 10^{-19}$ C)

$\left[1.12 \times 10^{-18} \text{ N} \right]$

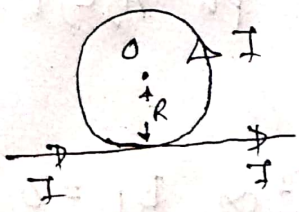
Find $\vec{B}$ at O in each case

4. a) 

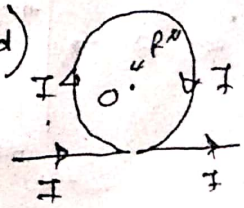
$\left[\frac{\mu_0}{4\pi} \frac{I}{R} (\pi + 1) \otimes \right]$

b) 

$\left[\frac{\mu_0 I \theta}{4\pi} \left(\frac{1}{a} - \frac{1}{b} \right) \otimes \right]$

c) 

$\left[\frac{\mu_0 I}{2\pi R} (\pi + 1) \odot \right]$

d) 

$\left[\frac{\mu_0 I}{2\pi R} (\pi - 1) \otimes \right]$

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5. A square loop carry a current and magnetic field at its centre is 'B_1'. A circular loop of same perimeter contains same current and magnetic field at its centre is 'B_2'. Find $\frac{B_1}{B_2}$ $\left(\frac{8\sqrt{2}}{\pi^2}\right)$

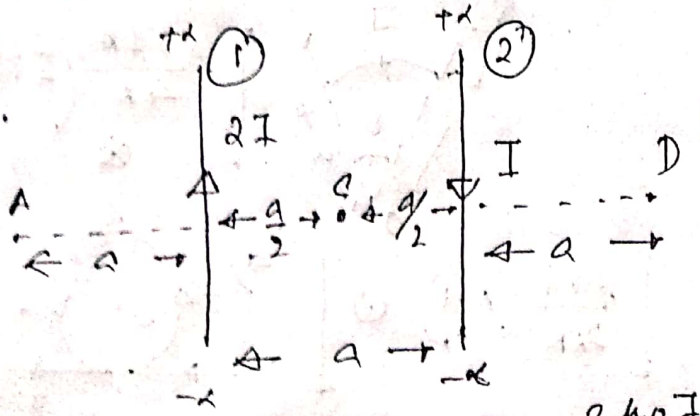
6. orbital radius of an electron of charge 'e' is 'r'. Its angular velocity is 'ω'. Find magnetic field developed at centre.

$\left(B = \frac{\mu_0 \omega e}{4\pi r}\right)$

7. A long wire carrying a steady current is bent into a circular loop of one turn. The magnetic field at the centre of the loop is 'B'. It is then bent into a circular coil of 'n' turns. Find magnetic field at the centre of this coil of n turns. $[n^2 B]$

8. A proton of KE 1 MeV is moving in a circular path in a uniform magnetic field. Find KE of an 'α' particle moving in the same magnetic field in same circle (same radius) [1 MeV]

9.



Find magnetic field $\vec{B}$ at pts A, C, D.

$\left[B_A = \frac{3}{4}\frac{\mu_0 I}{\pi a} \odot,\ B_C = \frac{3\mu_0 I}{\pi a} \otimes,\ B_D = 0\right]$

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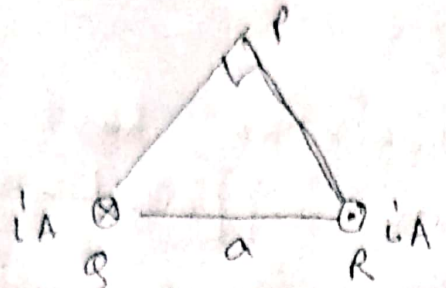
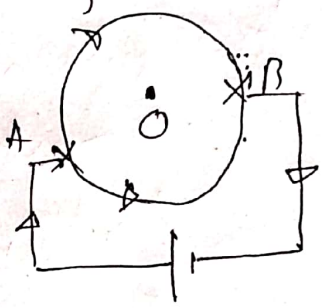
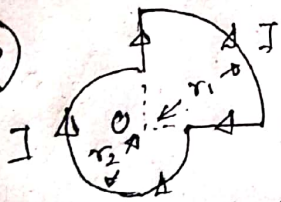
10. 

The fig. shows a right angled isosceles triangle PQR of base QR = 'a' m. A current of iA is flowing downwards along a thin straight wire cutting the plane of paper normally at Q. A similar wire at R carries an equal current iA, flowing upward at 'R'. Find magnitude and direction of magnetic field $\vec{B}$ at pt P. [wires are infinitely long & $\frac{\mu_0}{4\pi} = 10^{-7}$ T m A^{-1}]

[Ans $B = \underbrace{4\times10^{-7}\times\frac{i}{a}}$ T & $\vec{B}$ is in the plane of paper directed perpendicularly towards QR]

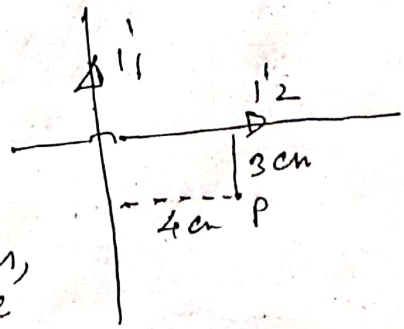
11. A battery is connected across two pts
a) A & B of a uniform circular conductor. [A & B are any two pts on the circular conductor] Prove that magnetic field at centre O is zero



b) 

Find B_O = ? [$B_O = \frac{\mu_0 I}{8}\left(\frac{1}{r_1}+\frac{3}{r_2}\right)$ ⊗ at O]

12. Two infinitely long insulated wires are kept ⊥ to each other & carry currents $i_1 = 2A$ & $i_2 = 1.5A$. Find magnitude and direction of magnetic field at P. [2×10^{-5} T, ⊥ to plane of paper, pointing downward ⊗] Ans.



If direction of current be reversed in one of the wires, what would be the magnitude of field at P [0].

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